



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)

EXAMINATIONS - DECEMBER 2022

SUBJECT : ECE 2152 - COMPUTER ORGANIZATION AND ARCHITECTURE

COMPUTER ORGANIZATION AND ARCHITECTURE [ECE 2152]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) For the expression $X = (A-B/C)/(D+E*F - G/H)$, write the program for register based, accumulator based and stack based machine. Assume source and destination operands are memory locations. (5)
- 1B) With the 12 bit instruction length and 4 bit address field, verify the possibility to have fourteen 2- address, thirty 1-address and twenty four 0-address instructions. (3)
- 1C) With the help of a block diagram explain the working of stack-based processor. (2)
- 2A) Explain the working of carry look ahead adder (CLA) circuit. Using 4 bit CLA as building block design a 16-bit CLA. Also, calculate worst case add time of 16-bit CLA. (5)
- 2B) Design a fast 8 x 8 multiplier using 4 X 4 non additive multipliers and Wallace tree adders. (3)
- 2C) List the disadvantages of array multipliers. How is it overcome by Booth's and Modified Booth's algorithms? Explain. (2)
- 3A) Design a microprogrammed control unit for Booth's multiplier by giving the RTL code. Explain the control memory organization of Booth's multiplier with a neat diagram. (5)
- 3B) Build the hardware to implement each of the following register transfers:
i) If $Q[1]=0$ AND $Q[0]=1$ then $A \leftarrow A+1$ else $A \leftarrow A-1$
ii) If $Q[1:0]=01$ OR $Q[1:0]=10$ then $A \leftarrow B$
Assume A, B and Q are 4-bit registers. (3)
- 3C) Convert the following decimal numbers to 32-bit IEEE floating point representation. Write the result in hexadecimal format.
a) 45.125
b) 0.0125 (2)
- 4A) Distinguish the difference between maskable and non-maskable interrupts. Explain with neat diagrams, software polled approach for connecting multiple interrupts to a single interrupt line. List its advantages and disadvantages. (5)
- 4B) A computer system needs a 1K X 8 RAM. Assuming 256 X 4 RAM chips are available, show how to build required RAM. Draw a neat logic diagram of your implementation. (3)
- 4C) A certain memory system has a 32MB main memory and a 64 KB Cache. Block are 16 bytes in sizes. Show the main memory address format to determine the length of TAG, Cache line number and WORD fields for the Direct mapping technique. (2)

- 5A) What is Superscalar architecture? Explain in detail the characteristic features of superscalar architecture. (5)
- 5B) What is CISC architecture? Explain its characteristic features. (3)
- 5C) A four-stage pipeline has stage delays as 40, 60, 90, and 70 ns respectively. Given latch delay is 5 ns.
Calculate, (2)
- i) Pipeline cycle time
 - ii) Non-pipeline execution time

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